

A Literal Counterexample to the Displayed Daws Cotype Conjecture in arXiv:1901.07496

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, to Conjecture 2 *as literally displayed* in Gerasimova–Thom. The cited Daws conjecture uses a corrected exponent, so this packet does not claim to disprove Daws’s intended conjecture or Gerasimova–Thom Conjecture 1.

1 Source conjecture

The source is M. Gerasimova and A. Thom, *On the isometrisability of group actions on p -spaces*, arXiv:1901.07496v2 [1]. On printed page 7 it defines $B_p(\Gamma)$ using coefficients of isometric representations on q -spaces and then displays:

Conjecture 2 (Daws). The space $B_p(\Gamma)$ has cotype q , where $\frac{1}{p} + \frac{1}{q} = 1$.

No range for p is included in the displayed statement. The complete source context is shown below.

$T_1(\Gamma) \subset B(\Gamma)$, where $B(\Gamma)$ is a space of coefficients of unitary representations. Secondly, using the fact that $B(\Gamma)$ has cotype 2, one can prove that $T_1(\Gamma) \subset \ell^2(\Gamma)$. The possible strategy to prove Conjecture 1 is to use $B_p(\Gamma)$ instead of $B(\Gamma)$, where $B_p(\Gamma)$ is a space of coefficients of isometric representation on q -spaces, see [12] for details:

$$B_p(\Gamma) = \{f: \Gamma \rightarrow \mathbb{C} \mid f \text{ is a coefficient of some } (\pi, E) \in \text{Rep}_q(\Gamma)\}.$$

The first step of the proof is quite similar to the classical case, see for example [1] or [10]. Thus we have,

PROPOSITION 1. *If Γ is $(2, p)$ -isometrisable, then $T_1(\Gamma) \subset B_q(\Gamma)$.*

To obtain $T_1(\Gamma) \subset \ell^q(\Gamma)$ one would need to use the cotype argument. Maybe the following conjecture that appeared in [3] holds:

CONJECTURE 2 (Daws). *The space $B_p(\Gamma)$ has cotype q , where $\frac{1}{p} + \frac{1}{q} = 1$.*

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2 Definitions

A Banach space X has (Rademacher) cotype r with constant C if, for every finite family $x_1, \dots, x_n \in X$,

$$\left(\sum_{j=1}^n \|x_j\|^r \right)^{1/r} \leq C \left(\mathbb{E}_\varepsilon \left\| \sum_{j=1}^n \varepsilon_j x_j \right\|^2 \right)^{1/2},$$

where the ε_j are independent random signs. Standard terminology only assigns cotype exponents in $[2, \infty)$; the displayed inequality also makes the obstruction below meaningful for $r < 2$.

The source's coefficient space $B_p(\Gamma)$ contains the coefficient of every isometric representation on a q -space, with $q = p/(p-1)$. In particular the trivial one-dimensional representation on $\mathbb{C}$ is allowed.

3 Idea of the proof

The obstruction is independent of group representation theory. Repeating one nonzero vector n times makes the left side of the cotype inequality grow as $n^{1/r}$, whereas the Rademacher square mean on the right grows only as $n^{1/2}$. Thus no nonzero Banach space can have cotype below 2. For $p > 2$ the conjugate exponent in the displayed conjecture is below 2, and $B_p(\Gamma)$ is nonzero because it contains the constant coefficient 1.

4 Counterexample

Lemma 1. *No nonzero Banach space has Rademacher cotype $r < 2$.*

Proof. Let $0 \neq x \in X$ and apply the cotype inequality to $x_1 = \dots = x_n = x$. Independence and $\mathbb{E}|\sum_{j=1}^n \varepsilon_j|^2 = n$ give

$$n^{1/r} \|x\| \leq C \sqrt{n} \|x\|.$$

Hence $n^{1/r-1/2} \leq C$ for every n , impossible when $r < 2$. □

Theorem 2. *Conjecture 2 of arXiv:1901.07496v2 is false as literally displayed. Explicitly, take*

$$p = 3, \quad q = \frac{3}{2}, \quad \Gamma = \{e\}.$$

Then $B_3(\Gamma) \cong \mathbb{C}$ is nonzero and does not have cotype $3/2$. More generally, the displayed conclusion fails for every $2 < p < \infty$ and every discrete group Γ .

Proof. The trivial representation of Γ on the one-dimensional q -space $\mathbb{C}$ is isometric, and its coefficient obtained from $x = 1$ and the identity functional is the constant function 1. Thus $B_p(\Gamma) \neq \{0\}$. If $p > 2$, its conjugate exponent $q = p/(p-1)$ satisfies $q < 2$, so Lemma 1 rules out cotype q . For the trivial group, every coefficient is a scalar function on one point, hence $B_3(\{e\}) \cong \mathbb{C}$. □

5 Convention diagnosis and limitations

The cited paper of Daws does *not* state the unrestricted formula in the displayed source conjecture. In Section 8.3, immediately after Proposition 8.8, Daws writes that Runde's algebra $B_p(G)$ is conjectured to have cotype

$$\max(p, p')$$

[2, p. 74]. Moreover, Daws explicitly warns that his Herz convention sometimes interchanges p and p' relative to Runde’s convention. These two details explain the mismatch. The exponent $\max(p, p')$ is always at least 2 and survives the elementary obstruction above.

Consequently, Theorem 2 resolves only the literal unrestricted displayed Conjecture 2. It does not settle the corrected Daws conjecture, and it gives no counterexample to Conjecture 1 concerning $(2, p)$ -isometrisability and Littlewood functions.

6 Verification and novelty check

The proof uses only the defining cotype inequality and the trivial one-dimensional coefficient; there is no numerical or computer-assisted step. A separate proof audit is included as `verification.md`.

A bounded search through 13 August 2026 covered the run’s four lightweight indexes, the current source and bibliography, the exact displayed phrase, the title and citations of arXiv:1901.07496, and arXiv-oriented searches for $B_p(G)$ with cotype. No later paper explicitly flagging this displayed range error was found. The original cited Daws paper was inspected directly and supplies the corrected $\max(p, p')$ formulation. Novelty confidence is therefore limited to the explicit diagnosis and counterexample to the source statement; the intended corrected conjecture is known from Daws and remains open.

7 Human-review recommendation

The mathematical counterexample is elementary. Review should focus on editorial scope: whether the displayed statement is meant to inherit an unstated range or Daws’s alternate convention. The packet should remain labelled “literal counterexample” and must not be reported as a disproof of the corrected Daws conjecture.

References

- [1] M. Gerasimova and A. Thom, *On the isometrisability of group actions on p -spaces*, arXiv:1901.07496v2, 2020.
- [2] M. Daws, *p -operator spaces and Figà–Talamanca–Herz algebras*, Journal of Operator Theory 63 (2010), 47–83; arXiv:math/0611833.